

Twenty-fifth day of class - 11/20

Fun fact: Suppose that n is a positive integer and n can be written as the sum of two rational cubes in only finitely many ways. Then n is a cube or n is twice a cube.

The curve $x^3 + y^3 = nz^3$ can be transformed into the elliptic curve $E : y^2 = x^3 - 432n^2$ and it is rare for this curve to have a point $P \in E(\mathbb{Q})$ that has order greater than 1 but not infinite order.

Last time: Elliptic curve primality proving.

Today: The Birch and Swinnerton-Dyer conjecture.

In the the fall of 1958, Bryan Birch and Peter Swinnerton-Dyer did some computations on the EDSAC at the University of Cambridge.

If E is an elliptic curve, let $N_p(E) = |E(\mathbb{F}_p)|$. By Hasse's theorem, we can write

$$N_p(E) = p + 1 - a_p(E)$$

and $|a_p(E)| \leq 2\sqrt{p}$.

For several different elliptic curve E , they computed

$$\prod_{p \leq x} \frac{N_p}{p}.$$

The data they gathered suggested that

$$\prod_{p \leq x} \frac{N_p}{p} \approx C(\log x)^r$$

where C is a nonzero constant, and r is the rank of $E(\mathbb{Q})$! Another way of saying this is that the presence of infinitely many rational points on $E(\mathbb{Q})$ “skews” the number N_p to make it likely to be larger than $p + 1$ more often.

This skew is fairly subtle. In particular, given some interval $[\alpha, \beta] \subseteq [-2, 2]$, the proportion of primes p for which

$$\frac{a_p(E)}{\sqrt{p}} \in [\alpha, \beta]$$

doesn't depend on the rank. (It only depends on whether the curve has CM or not.)

Example:

Let $E_1 : y^2 + y = x^3 - 7x + 6$, and $E_2 : y^2 + y = x^3 - 7x + 7$.

p	$N_p(E_1)$	$N_p(E_2)$
2	5	1
3	7	1
5	10	5
7	12	9
11	18	15
13	18	17
17	22	20
19	27	15
23	30	24

The curve E_1 has rank 3, and the curve E_2 has rank zero.

Formalizing Birch and Swinnerton-Dyer's observation in a useful way was difficult (and Birch and Swinnerton-Dyer had some assistance from Harold Davenport).

Definitions

Let $E/\mathbb{Q}$ be an elliptic curve and choose the minimal model for it. For each prime p for which E has good reduction, we have already defined $a_p(E) = p + 1 - |E(\mathbb{F}_p)|$. If E does not have good reduction modulo p , then E either has a node (self-intersection) or a cusp modulo p . If $E/\mathbb{F}_p$ has a cusp, define $a_p(E) = 0$. If it has a node, then at this singular point P , there are two different tangent lines to E , and the slopes of those tangent lines either both lie in $\mathbb{F}_p$, or neither lie in $\mathbb{F}_p$.

In the first case, define $a_p(E) = 1$ and in the second case, define $a_p(E) = -1$.

Example: Let $E : y^2 = x^3 - 16x + 16$ and $p = 37$. Modulo 37 we have the factorization $y^2 = (x + 3)(x + 17)^2$. The curve is singular, and the singular point is $x = -17 = 20$, $y = 0$.

$$\begin{aligned}
2y \frac{dy}{dx} &= 1(x + 17)^2 + 2(x + 3)(x + 17) \\
&= (x + 17) [(x + 17) + 2(x + 3)] = (x + 17)(3x + 23) \\
4y^2 \left(\frac{dy}{dx} \right)^2 &= (x + 17)^2 (3x + 23)^2 \\
4(x + 17)^2 (x + 3) \left(\frac{dy}{dx} \right)^2 &= (x + 17)^2 (3x + 23)^2 \\
\left(\frac{dy}{dx} \right)^2 &= \frac{(3x + 23)^2}{4(x + 3)}.
\end{aligned}$$

If we set $x = 20$, we get $\frac{dy}{dx}^2 \equiv 24 \pmod{37}$. However, there are no solutions to $z^2 \equiv 24 \pmod{37}$. Thus, $a_{37}(E) = -1$.

We have defined a number $a_p(E)$ for every single prime p . Now, define

$$a_{p^k}(E) = \begin{cases} a_p(E)a_{p^{k-1}}(E) - pa_{p^{k-2}}(E) & \text{if } p \text{ is a prime of good reduction and } k \geq 2 \\ a_p(E)^k & \text{if } p \text{ is a prime of bad reduction.} \end{cases}$$

Finally, if n is any positive integer with $n = \prod_{i=1}^k p_i^{e_i}$, define

$$a_n(E) = \prod_{i=1}^k a_{p_i^{e_i}}(E).$$

Definition. *The L -function of E is the series*

$$L(E, s) = \sum_{n=1}^{\infty} \frac{a_n(E)}{n^s}.$$

This series is like the function

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

This function, which may be the most studied function in all of mathematics, has deep connections to the distribution of prime numbers (and the Riemann hypothesis, one of the Millennium Prize Problems, is about the zeroes of this function).

The series for $\zeta(s)$ converges if s is any complex number $s = \sigma + it$ where $\sigma > 1$, but there are ways to extend the definition of $\zeta(s)$ to the entire complex plane.

The series for $L(E, s)$ is only guaranteed to converge (based on Hasse's bound on $a_p(E)$) for $s > 3/2$.

Conjecture 1: The function $L(E, s)$ can be extended to the entire complex plane.

Conjecture 2 (weak BSD): The function $L(E, s)$ has a zero at $s = 1$ with multiplicity equal to the rank of $E(\mathbb{Q})$. In other words, if we look at the Taylor series expansion of $L(E, s)$ at $s = 1$, we get

$$L(E, s) = c_0 + c_1(s - 1) + \frac{c_2}{2!}(s - 1)^2 + \dots$$

We are saying that $c_k = 0$ for $k < r$ and $c_r \neq 0$.

In particular, $r > 0$ if and only if $L(E, 1) = 0$.

Prize: Weak BSD is one of the 7 Millennium prize problems. If someone were to solve it, they would be eligible for a prize of 1 million dollars.

Conjecture 3 (strong BSD): There is an explicit formula for c_r , the first nonzero coefficient in the Taylor series expansion.

This formula involves a lot of things about the arithmetic of E :

- The real period of E

This is the number ω_1 where $E(\mathbb{C}) = \mathbb{C}/\Lambda$ and Λ is generated by $\omega_1 \in \mathbb{R}$ and $\omega_2 \in \mathbb{C}$.

- For each prime p for which E has bad reduction, there is a “Tamagawa fudge factor” $c_p(E)$. This is a positive integer, which says something about the index of the non-singular points modulo powers of p .

- The canonical heights $\hat{h}$ of the generators of $E(\mathbb{Q})$,

The canonical height of a point P is defined to be $\hat{h}(P) = \lim_{n \rightarrow \infty} \frac{h(2^n P)}{4^n}$. This function has the property that $\hat{h}(2P) = 4\hat{h}(P)$, and also that $\hat{h}(P + Q) + \hat{h}(P - Q) = 2\hat{h}(P) + 2\hat{h}(Q)$.

- The order of the torsion subgroup of E , and

- The order of the Shafarevich-Tate group.

What’s that?

Defn: A homogeneous space for a curve E is a genus one curve C and a “natural action” of E on C . For example, when we compute ranks, we work with curves

$$N^2 = dM^4 + aM^2e^2 + \frac{b}{d}e^4.$$

These are homogeneous spaces for E .

It turns out that there is a group structure on the set of homogeneous spaces. If G is the set of homogeneous spaces, there are two important subgroups of G .

Let K be the subgroup of homogeneous spaces that have rational points on them (when we work with them projectively).

Let H be the subgroup of homogeneous spaces which have points on them modulo n for every positive integer n . Note that K is a subgroup of H .

When $H = K$, our method of trying to determine $\text{im } \alpha$ and $\text{im } \bar{\alpha}$ works. Each curve C_d either has a point on it, or there is a modular arithmetic reason why C_d cannot have a point on it.

Define the Shafarevich-Tate group $\text{III}(E/\mathbb{Q})$ to be the quotient group K/H .

Conjecture: For any elliptic curve $E/\mathbb{Q}$, the Shafarevich-Tate group is finite.

John Tate said the following about BSD.

“This remarkable conjecture relates the behavior of a function L , at a point where it is not at present known to be defined, to the order of a group III , which is not known to be finite.”

Example of BSD

Let $E : y^2 = x^3 + 17x$. We can compute $L(E, 1) \approx 3.6523718207$. This is not difficult. Since $L(E, 1) \neq 0$, BSD predicts that the rank of $E(\mathbb{Q})$ is zero. This is in fact true.

The curve E is singular mod 2 and mod 17 and at each of these primes it has a cusp. Thus, $a_2(E) = a_{17}(E) = 0$.

The real period of E is $\Omega = 1.82618591\dots$. The strong BSD conjecture states in this case that

$$L(E, 1) = \frac{\Omega \left(\prod_{p|N(E)} c_p(E) \right) |\text{III}(E/\mathbb{Q})|}{|E(\mathbb{Q})_{\text{tors}}|^2}.$$

We have $c_2(E) = 1$ and $c_{17}(E) = 2$. We have $|E(\mathbb{Q})_{\text{tors}}| = 2$ and so the right hand side is

$$0.913092955 |\text{III}(E/\mathbb{Q})|$$

So if the strong BSD is true, then we expect

$$|\text{III}(E/\mathbb{Q})| = \frac{L(E, 1)}{0.913092955} \approx 4.00000.$$

That is, $|\text{III}(E/\mathbb{Q})| = 4$. In fact, we know that $|\text{III}(E/\mathbb{Q})| \geq 4$ because there are 16 possible d that could be in the image of α or $\bar{\alpha}$, and we know that there are only four with rational points.

What do we know about BSD?

The first significant partial result about BSD came from Andrew Wiles's PhD thesis (co-authored with his advisor John Coates).

Theorem (1977): If $E/\mathbb{Q}$ is an elliptic curve with CM, then $|E(\mathbb{Q})| = \infty$ implies that $L(E, 1) = 0$.

For CM curves, it was already known that $L(E, s)$ has an analytic continuation.

In 1988, a number of mathematicians succeeded in generalizing Wiles's work. and in particular showing that if $E/\mathbb{Q}$ is a “modular” elliptic curve,

Theorem (1988): If $E/\mathbb{Q}$ is a modular elliptic curve, then $L(E, 1) \neq 0$ implies that $E(\mathbb{Q})$ has rank zero, and $L(E, 1) = 0$ but $L'(E, 1) \neq 0$, then $E(\mathbb{Q})$ has rank one. (Also, in both cases $\text{III}(E/\mathbb{Q})$ is finite.)

The hypothesis that $E/\mathbb{Q}$ is modular implies that $L(E, s)$ has a continuation to all of $\mathbb{C}$.

Theorem (1999): Every elliptic curve $E/\mathbb{Q}$ is modular. In particular, $L(E, s)$ is infinitely differentiable at $s = 1$. (Hence, the weak BSD conjecture is well-defined.)

Definition. If $E/\mathbb{Q}$ is an elliptic curve, the conductor of E is a positive integer whose prime divisors are exactly the primes where E has bad reduction. If $p \geq 5$ and E has a node mod p , then $p|N(E)$ but $p^2 \nmid N(E)$. If E has a cusp mod p , then $p^2|N(E)$ but $p^3 \nmid N(E)$.

(There are special rules for the powers of 2 and 3 that divide $N(E)$. The power of 2 can be as high as 8 and the power of 3 can be as high as 5.)

In fact, the theorem says that more is true. Let $N(E)$ denote the conductor of E . Define

$$\Lambda(E, s) = N(E)^{s/2} \Gamma(s) L(E, s),$$

where $\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} dx$. Then,

$$\Lambda(E, 2-s) = \epsilon(E) \Lambda(E, s).$$

This gives a way of extending $L(E, s)$ to all complex numbers.

The number $\epsilon(E)$ is 1 or -1 and is called the *root number* of E . Because $\Gamma(1) = 1$, the Birch and Swinnerton-Dyer conjecture predicts that the rank of E is even if $\epsilon = 1$, and the rank of E is odd if $\epsilon = -1$.

Recall: If $p \equiv 1 \pmod{8}$ is prime and $E : y^2 = x^3 + px$, the book said that it is conjectured that the rank of E is always either 0 or 2. In this case, computing possibilities for the image of α and $\bar{\alpha}$ show that the rank of E is ≤ 2 . Also, for this curve E , $\epsilon(E) = 1$. Thus, the Birch and Swinnerton-Dyer conjecture predicts that the rank of E is even.

As of the formulas above, there is a very nice formula for $L(E, 1)$. It is

$$L(E, 1) = (1 + \epsilon(E)) \sum_{n=1}^{\infty} \frac{a_n(E)}{n} e^{-\frac{2\pi n}{\sqrt{N(E)}}}.$$

This formula converges quickly and can be used to determine if $L(E, 1)$ is zero or not, and then the Theorem from 1988 can be used.

(How quickly this series converges depends on how large $N(E)$ is. The larger $N(E)$, the more terms are needed.)

Note: The BSD conjecture, if true, sounds like it could be a useful way to compute the rank of an elliptic curve. We don't actually have a way to provably compute the order of vanishing of $L(E, s)$. The problem is that we can numerically compute any derivative $L^{(k)}(E, 1)$ to any amount of decimal accuracy we like, but that doesn't give us a way to prove that $L^{(k)}(E, 1) = 0$.